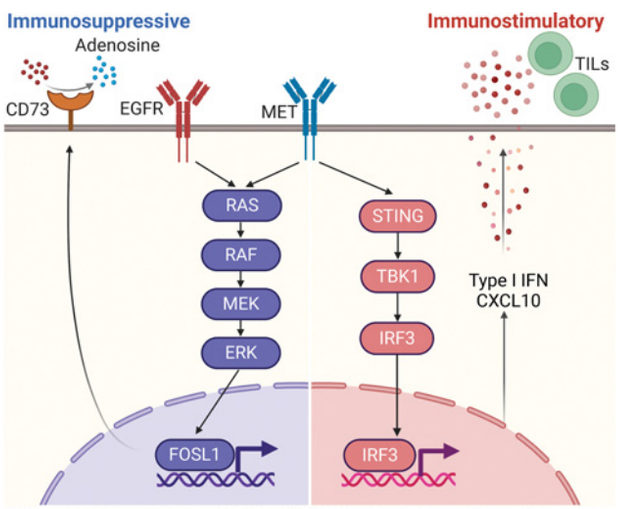
MET-Induced CD73 Restrains STING-Mediated Immunogenicity of EGFR-Mutant Lung Cancer

Ryohei Yoshida[1,2], Maria Saigi[1,3], Tetsuo Tani[1], Benjamin F. Springer[4], Hirofumi Shibata[1], Shunsuke Kitajima[1,5], Navin R. Mahadevan[1,6], Marco Campisi[1], William Kim[7], Yoshihisa Kobayashi[1,8], Tran C. Thai[1], Koji Haratani[1], Yurie Yamamoto[1,9], Shriram K. Sundararaman[1], Erik H. Knelson[1], Amir Vajdi[1], Israel Canadas[1,10], Ravindra Uppaluri[1], Cloud P. Paweletz[4,11], Juan J. Miret[4,11], Patrick H. Lizotte[1,4,11], Prafulla C. Gokhale[4], Pasi A. Jänne[1,11], and David A. Barbie[1,11]

ABSTRACT

Immunotherapy has shown limited efficacy in patients with *EGFR*-mutated lung cancer. Efforts to enhance the immunogenicity of *EGFR*-mutated lung cancer have been unsuccessful to date. Here, we discover that MET amplification, the most common mechanism of resistance to third-generation EGFR tyrosine kinase inhibitors (TKI), activates tumor cell STING, an emerging determinant of cancer immunogenicity (1). However, STING activation was restrained by ectonucleosidase CD73, which is induced in MET-amplified, EGFR-TKI–resistant cells. Systematic genomic analyses and cell line studies confirmed upregulation of CD73 in MET-amplified and MET-activated lung cancer contexts, which depends on coinduction of *FOSL1*. Pemetrexed (PEM), which is commonly used following EGFR-TKI treatment failure, was identified as an effective potentiator of STING-dependent TBK1-IRF3-STAT1 signaling in MET-amplified, EGFR-TKI–resistant cells. However, PEM treatment also induced adenosine production, which inhibited T-cell responsiveness. In an allogenic humanized mouse model, CD73 deletion enhanced immunogenicity of MET-amplified, EGFR-TKI–resistant cells, and PEM treatment promoted robust responses regardless of CD73 status. Using a physiologic antigen recognition model, inactivation of CD73 significantly increased antigen-specific $CD8^+$ T-cell immunogenicity following PEM treatment. These data reveal that combined PEM and CD73 inhibition can co-opt tumor cell STING induction in TKI-resistant *EGFR*-mutated lung cancers and promote immunogenicity.

Significance: MET amplification upregulates CD73 to suppress tumor cell STING induction and T-cell responsiveness in TKI-resistant, EGFR-mutated lung cancer, identifying a strategy to enhance immunogenicity and improve treatment.



CD73 upregulation in MET-amplified, EGFR TKI-resistant lung cancer impairs the activation of T cells by STING pathway signaling.

Introduction

EGFR-mutated non–small lung cancers (NSCLC) effectively respond to osimertinib and other third-generation EGFR tyrosine kinase inhibitors (EGFR-TKI); however, most patients present at an advanced stage and almost invariably develop acquired drug resistance (2). Approaches to combat EGFR-TKI resistance have centered around targeting downstream or parallel mitogenic signaling pathways such as MEK, MET, and IGF1R, which can compensate for and/or bypass EGFR-TKI inhibition (3). However, this strategy has been limited by the presence of drug-tolerant persister cells and the diversity of genetic and epigenetic mechanisms that can facilitate expansion of resistant clones (4).

[1]Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, Massachusetts. [2]Respiratory Center, Asahikawa Medical University, Hokkaido, Japan. [3]Department of Medical Oncology, Catalan Institute of Oncology (ICO), Germans Trias i Pujol Research Institute (IGTP), Badalona, Barcelona, Spain. [4]Experimental Therapeutics Core and Belfer Center for Applied Cancer Science, Dana-Farber Cancer Institute, Boston, Massachusetts. [5]Department of Cell Biology, Cancer Institute, Japanese Foundation for Cancer Research Tokyo, Japan. [6]Department of Pathology, Brigham and Women's Hospital, Boston, Massachusetts. [7]Jong Wook Kim Ph.D., University of California San Diego, School of Medicine, Moores Cancer Center. [8]Division of Molecular Pathology, National Cancer Center Research Institute, Tokyo, Japan. [9]Molecular Oncology and Therapeutics, Osaka Metropolitan University Graduate School of Medicine, Osaka, Japan. [10]Blood Cell Development and Function Program, Fox Chase Cancer Center, Philadelphia, Pennsylvania. [11]Belfer Center for Applied Cancer Science, Dana-Farber Cancer Institute, Boston, Massachusetts.

R. Yoshida and M. Saigi contributed as co-first authors to this article.

Corresponding Authors: David A. Barbie, Dana-Farber Cancer Institute, 450 Brookline Ave, LC4115, Boston, MA 02215. Phone: 617-632-6036; E-mail: david_barbie@dfci.harvard.edu; Pasi A Jänne, pasi_janne@dfci.harvard.edu; and Ryohei Yoshida, yryohei@asahikawa-med.ac.jp

Cancer Res 2022;82:4079–92

doi: 10.1158/0008-5472.CAN-22-0770

This open access article is distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license.

©2022 The Authors; Published by the American Association for Cancer Research

MET amplification and/or overexpression occurs in up to 25% of patients with NSCLCs overall (5, 6). It represents one of the most common resistance mechanisms to osimertinib, and along with MET exon 14 mutation represents a distinct targetable subtype of lung adenocarcinoma (7, 8). Several clinical trials examining the efficacy of combination therapy with MET inhibitors for *EGFR*-mutated NSCLCs are currently underway. In these trials, adverse events were found to be more severe than with EGFR-TKIs alone, although within acceptable limits. Combination therapy with the MET inhibitor savolitinib has advanced into a phase III trial and improves response of osimertinib resistant patients, but median progression-free survival in the phase II study was only 5.5 months, highlighting the limited durable activity of directly targeting this compensatory signaling pathway (9–11).

When T-cell immunity can be successfully engaged against lung cancer, responses are substantially more durable than for targeted therapies (12, 13). However, studies of immune checkpoint blockade (ICB) have revealed that patients with *EGFR*-mutated lung cancer rarely ever respond to immunotherapy, despite its impressive overall success in other subsets of NSCLC (14). Low tumor mutational burden and PD-L1 expression, along with low rate of $CD8^+$ tumor-infiltrating T cells are considered to be reasons for the failure of response to ICB, but the underlying mechanisms remain unknown (15, 16). On the other hand, the oncogenic activation of *MET* promotes immune escape by upregulating different transcripts such as *CD274* (encoding PD-L1) or *SOCS1*, among others, involved in immunosuppression (17).

Multiple studies have demonstrated an emerging role for innate immune signaling pathway dysregulation in *EGFR*-mutated lung cancer, and more specifically adaptive resistance to EGFR TKIs. These finding suggest that understanding and characterizing specific mechanisms of innate immune pathway dysregulation could uncover novel immunotherapeutic opportunities. For example, activation of STAT3 and NFκB signaling is implicated in promoting survival in response to EGFR-TKI treatment (18, 19). In addition, our group previously reported induction of STING, a critical downstream sensor of cytosolic double-stranded DNA (dsDNA) and a key component of the innate immunity, in the EGFR-TKI drug-tolerant persister cell state, together with secretion of specific proinflammatory cytokines (20). An EGFR-TKI–induced antiviral response involving upregulation of type I IFN signaling was also recently described by Gong and colleagues (21), highlighting the potential of such states to prime antitumor immunity. These data suggest that the residual cells that actually mediate acquired resistance to targeted therapy in *EGFR*-mutated tumors could be ideal targets for harnessing the innate immune response to promote tumor clearance.

However, induction of counterregulatory factors such as CD73, which generates the immunosuppressive metabolite adenosine, may suppress tumor immunogenicity (22), particularly in *EGFR*-mutated NSCLC (23, 24). Human CD73 is encoded by *NT5E*, which is located on the long arm (q) of chromosome 6. The canonical pathway leading to extracellular adenosine production involves the degradation of extracellular ATP into AMP and subsequently into adenosine, by the action of membrane-bound ectonucleotidases CD39 and CD73, respectively (25, 26). Extracellular adenosine (eADO) can signal through four different adenosine receptors, designated A1, A2a, A2b, and A3, dampening the immune response by suppressing immune effector cell function (22). More specifically, an increase of eADO, mainly through the adenosine A2A receptor signaling, impairs T-cell cytotoxicity and natural killer cell function and induces the suppression of antigen-presenting cells (27, 28). All of these molecules, among others, contribute to adenosinergic signaling in the tumor microenvironment, and their overexpression has been associated with metastatic disease and worse clinical outcomes in various tumor types, including NSCLC (29–31).

In this study, we sought to characterize STING expression in established EGFR-TKI–resistant lung cancer cells, discovering that its upregulation is linked to MET amplification. However, while examining antigen-specific T-cell activation in this MET-activated cellular context, we discovered a key role for CD73 in restraining STING-driven immunogenicity, prompting us to examine the potential of activating STING and inhibiting CD73 to overcome this barrier.

Materials and Methods

Cell lines

HEK293T, H1650, H441, and H3255 cells were purchased from ATCC. H23, H1975, and H820 cells were originally obtained from the Broad Institute, and H69 and H69M were originally obtained from the laboratory of Dr. Joan Albanell and were authenticated by short tandem repeat genotyping. HCC827, HCC827GR6, HCC827EPR, PC9, HCC2935, HCC2279, HCC4006, DFCI-202, EBC1, H596, and DFCI-633 cells were kindly provided by Dr. P.A. Jänne at Dana-Farber Cancer Institute (DFCI), Boston, MA. Jurkat76 cell line was provided by M.H.M Heemskerk at Leiden University Medical Center, Leiden, the Netherlands. HEK293T cell was cultured in DMEM (Thermo Fisher Scientific, catalog no. 11965-118), 10% FBS (GeminiBio products, catalog no. 100-106), and 1% penicillin/streptomycin (Gibco, #15140-122). H1650, H3255, H1975, H820, HCC827, HCC827GR6, HCC827EPR, PC9, HCC2935, HCC2279, HCC4006, H69, H69M, H441, EBC1, H596, H23, DFCI-202, DFCI-633, and Jurkat76 cell lines were cultured at 37°C in 5% CO_2 using RPMI1640 (Thermo Fisher Scientific, catalog no.11875-119), 10% FBS and 1% penicillin/streptomycin. *Mycoplasma* infection was regularly checked by PCR using the condition media derived from each cell line. The sequences of the primers used for checking *Mycoplasma* infection are listed in Supplementary Table S1.

Reagents and treatments

The following reagents were used: osimertinib (Selleckchem, catalog no. S7297), savolitinib (Selleckchem, catalog no. S7674), pemetrexed (Selleckchem, catalog no. S5971), docetaxel (Selleckchem, catalog no. S1148), vinorelbine (Selleckchem, catalog no. S4269), etoposide (Selleckchem, catalog no. S1225), Olaparib (Selleckchem, catalog no. S1060), barasertib (Selleckchem, catalog no. S1147), cisplatin (Selleckchem, catalog no. S1166), ruxolitinib (Selleckchem, catalog no. S1378), trametinib (Selleckchem, catalog no. S2673), crizotinib (Selleckchem, catalog no. S1068), AB680 (MedChemExpress, catalog no. HY-125286), adenosine (Sigma-Aldrich, catalog no. A9251), AMP (Sigma-Aldrich, catalog no. A1752), APCP (Sigma-Aldrich, catalog no. M8386), Recombinant Human HGF (R&D Systems, catalog no. 294-HG-005/CF).

Patient samples and IHC staining

Patients with NSCLC with *EGFR* mutation were identified through the DFCI PROFILE database. Formalin-fixed paraffin-embedded with 4-μm-thick were stained at the Division of Pathology in Brigham & Women's Hospital Pathology Core. Staining for the following antibodies was performed on BOND-III, the fully automated IHC and ISH stainer (Leica Biosystems): STING (Cell Signaling Technology, catalog no. 13647, dilution 1:50), CD73 (Cell Signaling Technology, catalog no. 13160, dilution 1:50), c-MET (Abcam, catalog no. ab243930, dilution 1:100). Poly-HRP IgG reagent from Bond Polymer Refine Detection Kit DC9800 was used to bind rabbit antibody. Expression

levels of STING, CD73, and c-MET staining in tumor cells were visually evaluated by a pathologist who was blinded to other data. We used a IHC score system (H-score) that was calculated by multiplying the membranous intensity score in tumor cells (0, absent; 1, weak; 2, moderate; 3, strong) by the percentage of stained cells (0%–100%) to yield a value of 0–300 m and classified into high versus low expression at the median cut-off point (32).

CRISPR/Cas9 system and lentiviral infection

Target sequences for CRISPR interference were designed using the single-guide RNA (sgRNA) designer (http://portals.broadinstitute.org/gpp/public/analysis-tools/sgrna-design). A nontargeting sgRNA from the Gecko library v2 was used as a scramble sgRNA. sgRNA target sequences are listed in Supplementary Table S1.

A total of 3×10^6 HEK293T cells were plated onto a 60-mm dish and transfected using X-tremeGENE HP DNA Transfection Reagent (Roche, catalog no. 06366236001) with 1 μg of lentivirus-based expression vectors together with 1 μg of pCMV-dR8.91 and 1 μg of pCMV-VSV-G. After 48 hours incubation, the media containing lentivirus particles were collected, passed through a 0.45 μmol/L filter, and concentrated using Lenti-X Concentrator (Clontech, catalog no. 631231). The constructs of TCR-LMP2B were purchased from VectorBuilder company. For selection of virally infected cells, 0.5–2 μg/mL of puromycin (pCRISPR-v2 sgRNAs, plx307-hMET) or 1.5–8 μg/mL of blasticidin (plx304-NanoLuc, plx304-hSTING, plx304-CD73, or plx304-hFRA1) was used 24 hours postinfection.

qRT-PCR

RNA extraction was performed using RNeasy Mini Kit (Qiagen, catalog no. 74106). RNA samples (1 μg) were reverse transcribed using SuperScript III First-Strand Synthesis SuperMix (Thermo Fisher Scientific, catalog no. 1683483). qRT-PCR was performed using Power SYBR Green PCR Master Mix (Thermo Fisher Scientific, catalog no. 4367659). The sequences of the primers used for qRT-PCR are listed in Supplementary Table S1.

Immunoblotting

Cells were lysed in RIPA lysis and extraction buffer (Thermo Fisher Scientific, catalog no. 89900) containing 1× protease inhibitors (Roche, catalog no. 11-836-145-001) and phosphatase inhibitor (50 μmol/L NaF and 100 μmol/L Na_3VO_4). Immunoblotting was performed as described previously (33) using antibodies that specifically recognize MET (Cell Signaling Technology, catalog no. 8198), Phospho-MET (Cell Signaling Technology, catalog no. 3077), STING (Cell Signaling Technology, catalog no. 13647), CD73 (Cell Signaling Technology, catalog no. 13160), TBK1 (Cell Signaling Technology, catalog no. 3013), Phospho-TBK1 (Cell Signaling Technology, catalog no. 5483), IRF3 (Cell Signaling Technology, catalog no. 11904), phospho-IRF3 (Cell Signaling Technology, catalog no. 4947), STAT1 (Cell Signaling Technology, catalog no. 9172), phosphor-STAT1 (Cell Signaling Technology, catalog no. 9167), ERK (Cell Signaling Technology, catalog no. 9107), phosphor-ERK (Cell Signaling Technology, catalog no. 4370), AKT (Cell Signaling Technology, catalog no. 9272), phosphor-AKT (Cell Signaling Technology, catalog no. 4060), FRA1 (Abcam, catalog no. 124722), phospho-FRA1 (Cell Signaling Technology, catalog no. 5841), cGAS(Cell Signaling Technology, catalog no. 15102), and β-actin (Cell Signaling Technology, catalog no. 3700). Secondary antibodies were from LI-COR Bioscience: IRDye 680LT Goat anti-Mouse IgG (#926-68020), IRDye 800CW Goat anti-Rabbit IgG (#926-32211). Imaging of blots was performed using LI-COR Odyssey system.

Flow cytometry assay

Cells in culture were examined for surface expression by flow cytometry with BD Biosciences LSRFortessa and the following antibodies: anti-HLA-A,B,C (BioLegend, catalog no. 311436), anti-CD73 (BioLegend, catalog no. 344004), anti-CD39 (R&D Systems, catalog no. FAB4397A), anti-PD-L1 (BioLegend, catalog no. 329718), anti-CD45 (BioLegend, catalog no. 368511) or isotype IgG control antibodies (BioLegend, catalog no. 400268, BioLegend, catalog no. 400114, R&D Systems, catalog no. IC002A, BioLegend, catalog no. 400326, BioLegend, catalog no. 400122). LMP2 Tetramer-SSCSSCPLSK was purchased from MBL. Cells resuspended in 100 μL PBS containing 3% FBS were stained by each antibody. FlowJo v10 was used to perform analysis of flow cytometry raw data.

Cell viability assay

A total of 2,000–5,000 cells were seeded in 96-well plates, allowed to attach overnight, and then incubated with growth media containing drugs as indicated for 96 hours. Values of CellTiter-Glo (CTG) Luminescent Cell Viability assay (Promega) after 96 hours were normalized to vehicle-treated cells. Plates were read on a Tecan Infinite M200 Pro plate reader and analysis was performed using Prism7 (GraphPad Software). All conditions were tested in triplicate. The values represent the average of three technical replicates and a representative experiment from at least two independent experiments (biological replicates).

Adenosine assay

We treated each cell line with DNA-damaging compounds for 48 hours. Viable cells were replated after treatment and after 24 hours cell culture medium was replaced with medium containing 1% FBS. After 72 hours, 50 μmol/L AMP was added 30 minutes before collecting conditioned medium (CM) and then adenosine quantification was performed according to the instructions in Adenosine Assay Kit (Abcam, catalog no. ab211094). All values were normalized to CM containing 1% FBS.

dsDNA stimulation

A total of $2–5 \times 10^5$ cells were plated onto a 6-well plate and transfected using X-tremeGENE HP DNA Transfection Reagent (Roche, catalog no. 06366236001) with the indicated amount of poly (dA:dT; Invivogen, catalog no. tlrl-patn).

siRNA transfection

siRNAs targeting for MET (s8702, catalog no. 4390824) and negative control (catalog no. 4390843) were purchased from Thermo Fisher Scientific. Cells were transfected with 40 nmol/L of the siRNAs using Lipofectamine RNAiMAX Transfection Reagent (catalog no. 13778030) following the manufacturer's instructions, and collected after 48 hours culture.

ELISA

Human CXCL10 (R&D Systems, catalog no. DIP100), human IFNγ (R&D Systems, catalog no. DIF50C), and human IL2 (R&D Systems, catalog no. D2050) ELISAs were performed according to the manufacturer's instructions. Conditioned media from each cell lines was collected after 24 or 72 hours culture.

Murine and tumor implantation studies

Female NSG (NOD-*scid* IL2Rgamma null) mice, 3 weeks old were purchased from The Jackson Laboratory. Animals were acclimated for 5 days before initiation of the study. The study was conducted at DFCI

with the approval of the Institutional Animal Care and Use Committee in an Association for Assessment and Accreditation of Laboratory Animal Care International accredited vivarium.

Whole body radiation of 2 Gy was administered to the 4-week-old NSG mice using the X-Rad 225Cx Image Guided Biological Irradiator System (Precision X-Ray). Within 2 hours after radiation, mice were implanted intravenously with 35,000 human $CD34^+$ cells from a single donor cord blood (HumanCells Bioscience). Mice were monitored for human immune cell engraftment. At 13 weeks after human $CD34^+$ cell implantation, mice were randomized using Studylog software for $hCD45^+$ cell engraftment and implanted subcutaneously with 5×10^6 HCC827GR6 or HCC827GR6-CD73 knockout (KO) cells with 30% Matrigel (Thermo Fisher Scientific) in the hind-flank (Supplementary Table S2). Tumor growth monitoring was initiated 1 week after tumor cell implantation with $n = 8$/group. The tumor volume was calculated using the following formula: $(mm^3) = \text{length} \times \text{width} \times \text{width} \times 0.5$. On day 31 after implantation, $n = 5$ mice/xenograft were euthanized, and tumors collected fresh for FACS analysis and a small piece fixed in formalin for IHC. In the remaining $n = 3$/xenograft on days 35–39, mice were treated with pemetrexed (purchased from DFCI pharmacy), 100 mg/kg i.p. Once the treatment was completed, tumors were monitored at least twice a week. Data collection and analysis were not performed blind to the conditions of the experiments.

Fresh tumors were mechanically and enzymatically disaggregated in dissociation buffer consisting of RPMI (Life Technologies) +10% FBS (HyClone), 100 U/mL collagenase type IV (Life Technologies), and 50 μg/mL DNase I (Roche). Suspension was incubated at 37°C for 45 minutes and then further mechanically dissociated. Red blood cells were removed from samples using red blood cell lysis buffer (BioLegend). Samples were pelleted and then resuspended in fresh RPMI +10% FBS and strained through a 40 μm filter. Cells were incubated with the Live/Dead Zombie NIR (BioLegend) for 5 minutes the dark at room temperature. Fc receptors were blocked prior to surface antibody staining using 1:1 mix of Human and Mouse TruStain FcX Blocking Reagent (BioLegend). Cells were stained for 15 minutes on ice in the dark and washed 2× with PBS + 2% FBS. Cells were analyzed on a BD LSRFortessa with FACSDiva software (BD Biosciences). Data were analyzed using FlowJo software version 10.5.3. Antibodies were specific for the following markers: CD16(3G8) (BioLegend, catalog no. 302006), CD8(RPA-T8) (Thermo Fisher Scientific, catalog no. BDB560662), CD56 (GDC56) (BioLegend, catalog no. 318348), PD-L1 (29E.2A3) (BioLegend, catalog no. 329708), CD3 (UCHT1) (BioLegend, catalog no. 300424), HLA-DR (G46-6; Thermo Fisher Scientific, catalog no. BDB562804), CD45RO (UCHL1) (BioLegend, catalog no. 304237), CD15 (SSEA-1) (BioLegend, catalog no. 323028), CD19 (HIB19) (BioLegend, catalog no. 302243), CD45 (H130) (BioLegend, catalog no. 304050), CD4(RPA-T4) (BioLegend, catalog no. 300554), CD14 (M5E2) (BioLegend, catalog no. 301840).

Statistical analysis

Statistical significance was assessed using unpaired two-tailed Student *t* test, one-way ANOVA followed by Tukey *post hoc* test, or two-way ANOVA followed by Tukey *post hoc* test. *P* values less than 0.05 were considered significant. Asterisks used to indicate significance correspond with: *, $P < 0.05$; **, $P < 0.005$; ***, $P < 0.001$. Columns represent means ± SD. In one-way or two-way ANOVA followed by *post hoc* tests, we showed asterisks only in pairs of our interest. GraphPad Prism7 was used for all statistical analysis.

RNA sequencing for cancer cell lines analysis

We first compared between protein and mRNA levels of CD73/NT5E across cancer cells using quantitative proteomic and RNA sequencing (RNA-seq) data from Cancer Cell Line Encyclopedia (CCLE; refs. 37, 38) from DepMap portal (https://depmap.org/portal). For RNA-seq data, TPM (transcripts per million) values as defined by RSEM and $\log_2$ transformed using a pseudocount of 1 to avoid zeros. We also compared the CD73 protein levels to identify top associated CRISPR dependencies by using Chronos scores from DepMap 21Q4 release (39, 40). Reverse-phase protein array (RPPA) data were derived as described previously (34). We performed our analysis on 272 cancer cell lines from solid tumors, which overlapped among the three datasets. We calculated the degree of association between each feature using probabilistic models and Information-Theoretic metrics of association (IC, information coefficient). The statistical significance was estimated using an empirical permutation test (*P* values) and FDR after performing 1,000 permutations.

Data availability

The datasets generated during and/or analyzed in this study are available from the corresponding author upon reasonable request.

Study approval

All patients were consented to an Institutional Review Board (IRB)-approved protocol about specimen collection and correlation with clinical data (IRB#02-180). This study was performed in accordance with the Declaration of Helsinki.

Results

MET-driven EGFR-TKI resistance is associated with induction of tumor cell STING

We used the parental HCC827 lung cancer cell line that harbors an *EGFR*-activating mutation (deletion in exon19) and two isogenic resistant cell lines: HCC827GR6, which harbors *MET* amplification, and HCC827EPR, which harbors *EGFR* T790M mutation, to compare innate immune signaling in these two different contexts (**Fig. 1A**). STING was strongly upregulated in HCC827GR6 cells relative to HCC827 and HCC827EPR cells (**Fig. 1B**; Supplementary Fig. S1A). As expected, HCC827GR6 cells were sensitive to the MET-TKI savolitinib in combination with osimertinib (Supplementary Fig. S1B and S1C). Savolitinib and osimertinib treatment also reduced levels of STING protein in HCC827GR6 cells, suggesting the possibility that MET signaling might be responsible for maintaining STING expression in conjunction with EGFR (**Fig. 1C**).

To assess whether the relationship between elevated *MET* and tumor cell STING expression might be more broadly generalizable across *EGFR*-mutated lung cancer cell lines, we next analyzed transcriptomic data from the CCLE (**Fig. 1D**). Indeed, high STING (also known as *TMEM173*) and *MET* mRNA expression were significantly correlated with each other in this dataset ($R = 0.6495$, $P = 0.0223$). To validate this finding further, we used IHC to analyze tumor cell–specific levels of STING and MET protein, across a panel of 38 patient-derived *EGFR*-mutated NSCLC xenograft (PDX) samples, enriched for EGFR-TKI–resistant samples. High-intensity STING staining was found to be robustly associated with high MET protein levels in tumor cells ($P = 0.0193$, Fisher exact test; **Fig. 1E**; Supplementary Table S3), reinforcing that STING is upregulated in concert with MET in *EGFR*-mutated lung cancers.

To assess the causal relationship between MET signaling and regulation of STING expression in *EGFR*-mutated lung cancers more

directly, we next used CRISPR/CAS9 to delete *MET* specifically in HCC827GR6 or DFCI-202 cells, another model of *MET*-amplified *EGFR*-mutated lung cancer cells that upregulated STING (Supplementary Fig. S1D and S1E). *MET* deletion decreased STING levels (**Fig.1F**), which was especially apparent following generation of single-cell clones in HCC827GR6 cells. (**Fig. 1G**; Supplementary Fig. S1F). HCC827GR6 cells transfected with siRNA or different sgRNAs targeting *MET* also downregulated STING levels in accordance with the degree of *MET* suppression (Supplementary Fig. S1G and S1H). Conversely, exogenous overexpression of MET in HCC827 in multiple additional *EGFR*-mutated cell lines (H1975 and HCC4006) induced STING expression (**Fig. 1H**). Together, these findings demonstrate that *MET* amplification promotes expression of STING in *EGFR*-mutated lung cancer cells.

We next explored the downstream impact on cytoplasmic dsDNA sensing in *MET*-driven EGFR-TKI–resistant lung cancer cells. Transfection of HCC827GR6 cells with the dsDNA mimic poly (dA:dT) induced pTBK1, pIRF3, and pSTAT1, which was impaired following *MET* or *STING* deletion (**Fig. 1I**). In consonance with this observation, *MET* or *STING* deletion in HCC827GR6 cells also reduced poly (dA:dT) induced CXCL10 secretion, a potent T-cell chemokine activated downstream of IRF3 and STAT1 (**Fig. 1J** and **K**). Pathway activation was not substantially observed in the parental HCC827 cell line following poly (dA:dT) treatment (Supplementary Fig. S1I). We also observed similar results using the downstream cGAS product 2′3′ cGAMP, which preferentially activated STING signaling in HCC827GR6 cells as compared with HCC827cells (Supplementary Fig. S1J and S1K). Thus, activation of STING by oncogenic *MET* regulates downstream TBK1-IRF3-STAT1 signaling in EGFR-TKI–resistant lung cancer cells.

Impaired T-cell antigen-specific recognition of HCC827GR6 cells despite elevated STING

Tumor cell STING activation is an increasingly recognized determinant of effective antigen presentation because it can improve display of immunogenic peptides to CD8 T cells (35). To assess the impact of elevated STING on antigen-presenting function in *MET*-driven resistant *EGFR*-mutated lung cancer, we generated a model of antigen-specific recognition using HCC827 cells. Because HCC827 cells express the HLA A*11 allele, we established a model using a defined epitope (SSCSSCPLSK) from LMP2 of Epstein-Barr Virus, which binds HLA A*11 and for which a specific αβTCR has been defined (36). We introduced this T-cell receptor (TCR) into the J76 Jurkat T-cell line (37) and confirmed that these J76 TCR-LMP2B cells expressed the αβTCR relative to parental J76 cells (**Fig. 2A**). Next, we used the LMP2 Tetramer-SSCSSCPLSK to confirm that J76 TCR-LMP2B cells indeed express the SSCSSCPLSK/HLA A*11-specific TCR (**Fig. 2B**). JC76 TCR-LMP2B cells were utilized as effector cells (E), and HCC827 cells as target cells (T), which were loaded with the SSCSSCPLSK peptide. We confirmed that J76 TCR-LMP2B cells produced increasing levels of IFNγ in response to increasing E:T ratios, confirming antigen-specific recognition of HCC827 cells using this assay (**Fig. 2C**).

We therefore used this model to compare the immunogenicity of HCC827 and HCC827GR6 cells. In contrast to our expectation, TCR-LMP2B cells produced greater levels of IFNγ and IL2 in coculture with HCC827 cells as compared with HCC827GR6 cells (**Fig. 2D**). These data suggested that, despite elevated STING expression, coinduction of additional factors that suppress immunogenicity might be present in HCC827GR6 cells, which prompted us to further explore the potential mechanisms that could impair immunogenicity.

CD73 coactivation with STING in MET-driven EGFR-TKI–resistant cells

To understand additional factors that could influence tumor immunogenicity in the context of *MET* amplification, we first integrated RNA-seq data from NSCLC samples in The Cancer Genome Atlas (TCGA) and the CCLE database, and examined the top genes coexpressed with *MET* in both datasets (**Fig. 3A**). Twenty-eight genes were identified as candidates, including the immune suppressive factors *NT5E* (encoding CD73), which showed a strong correlation with MET expression in CCLE NSCLCs ($R = 0.5169$, $P < 0.0001$; **Fig. 3B**). We therefore examined the coexpression of these genes with *MET* specifically in *EGFR*-mutated lung adenocarcinoma in the CCLE, which identified a significant correlation between *MET* and *NT5E* expression ($R = 0.6932$, $P = 0.0124$), but not with *CD274* ($R = 0.3881$, $P = 0.2126$; **Fig. 3C**; Supplementary Fig. S2A). PD-L1 cell surface expression was also lower in HCC827GR6 cells, as compared with parental HCC827 and HCC827EPR cells, arguing against PD-L1 as a significant determinant of immune suppression in this setting (Supplementary Fig. S2B). Furthermore, *MET* was among the top CRISPR genetic codependencies with elevated *NT5E* mRNA expression and CD73 protein levels in the CCLE database, suggesting an important functional relationship between MET and CD73 (**Fig. 3D**). Given these findings, we therefore aimed to understand the contribution of *MET* amplification in the control of CD73 levels.

Indeed, similar to STING, CD73 was strongly upregulated in HCC827GR6 cells (**Fig. 3E**), and was specifically downregulated by savolitinib treatment, alone or in combination with osimertinib (**Fig. 3F**). In contrast, we did not observe expression of the upstream ectonucleosidase CD39 in either HCC827 or HCC827GR6 cells (**Fig. 3G**). We confirmed by IHC across the same panel of 38 *EGFR*-mutated NSCLC PDX samples that CD73 was robustly associated with MET protein levels in tumor cells ($P = 0.0029$, Fisher exact test; **Fig. 3H**; Supplementary Table S4). We therefore also measured CD73 levels on the cell surface and by immunoblot in both HCC827GR6 and DFCI-202 cells following MET CRISPR deletion, and observed downregulation compared with control scramble guides (**Figs. 3I** and **J**; Supplementary Fig. S2C and S2D). Furthermore, the same CRISPR-derived in HCC827GR6 single-cell clones, effectively depleted MET, STING, and CD73 (**Fig. 3K**; Supplementary Fig. S2E), as did MET siRNA treatment (Supplementary Fig. S2F). Taken together, these findings confirmed *MET*-driven coexpression of CD73 with STING in resistant *EGFR*-mutated lung cancers.

CD73 is more generally regulated by oncogenic activation of MET signaling in lung cancer cells

We next explored to what extent MET hyperactivation is directly involved in the upregulation of CD73, regardless of EGFR mutation status. We first examined previously described H69/H69M cell lines, a paired model of neuroendocrine to mesenchymal cell state switch induced by chronic hepatocyte growth factor (HGF) exposure, which activates both MET and STING expression in the H69M derivative (38, 39) We found that the levels of CD73 were strongly increased by immunoblot and flow cytometry in H69M cells compared with the parental H69 cells (**Fig. 4A** and **B**). We next selected a panel of lung cancer cell lines with varying *MET* status, including amplification (METamp), exon 14 skipping mutation (METex14), and wild type (**Fig. 4C**). Both EBC1 cells (METamp) and H441 cells (METamp, KRASmut) exhibited increased pMET and CD73 levels, whereas H596 cells, which harbor an *METex14* mutation, did not show elevated levels of CD73 at baseline but also had low levels of p-MET relative to these other cell lines. To elucidate whether the depletion of oncogenic MET

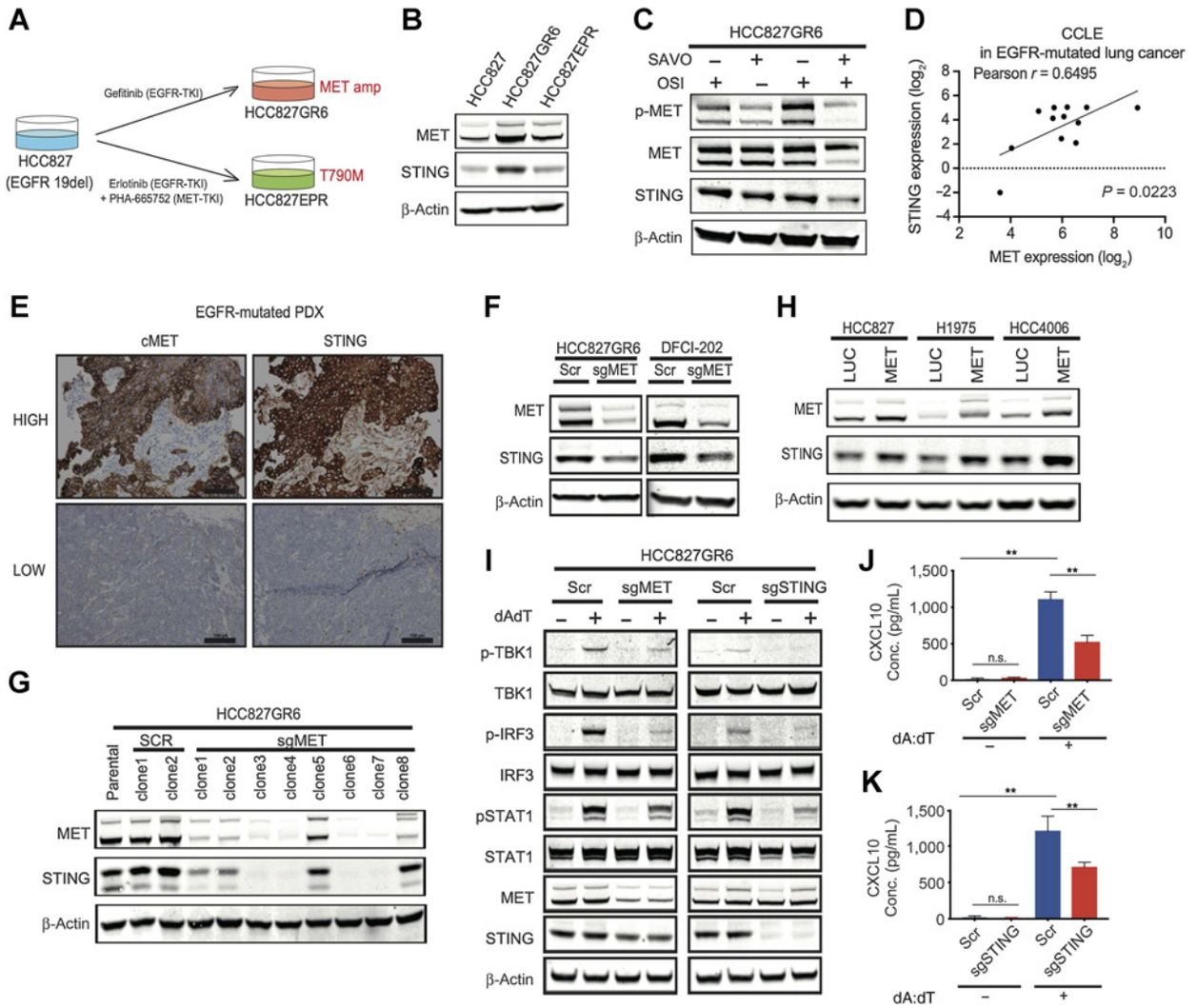


Figure 1.

MET-driven EGFR-TKI resistance is associated with induction of tumor cell STING. **A,** Schematic of EGFR-TKI–resistant cell lines and resistance mechanism using HCC827 cells. **B,** Immunoblot of the indicated proteins in HCC827, HCC827GR6, and HCC827EPR cell lines. **C,** Immunoblot of indicated proteins HCC827GR6 cell lines treated with 1 μmol/L osimertinib (OSI) and 1 μmol/L savolitinib (SAVO) for 72 hours. **D,** Gene expression correlation between MET and STING in EGFR-mutated lung cancer cell lines using CCLE. *R* squared and *P* values were calculated using Pearson correlation. **E,** Representative cMET and STING IHC images of EGFR-mutated PDX samples. **F,** Immunoblot of indicated proteins in HCC827GR6 or DFCI-202 cell lines with Scr or MET sgRNA. **G,** Immunoblot of single-cell cloning using HCC827GR6 cells with Scr or MET sgRNA. **H,** Immunoblot of the indicated proteins in overexpressing of MET in HCC827, H1975, and HCC4006 cell lines. **I,** Immunoblot of the indicated proteins in HCC827GR6 cells with Scramble (Scr) or MET sgRNA or STING sgRNA, and treated ± 1 μg/mL poly (dA:dT) for 4 hours. **J** and **K,** ELISA of human CXCL10 levels in CM derived from HCC827GR6 cells with Scramble or MET sgRNA or STING sgRNA, and treated ± 1μg/mL poly (dA:dT) for 24 hours. **, $P < 0.005$; n.s., nonsignificant.

reduced CD73 expression, we used CRISPR/CAS9 to generate *MET* KO in EBC1 cells. Compared with the control, the depletion of MET led to a strong reduction in the levels of total and cell surface CD73 expression (**Fig. 4D**), as previously observed in HCC827GR6. We also treated EBC1 cells with the MET inhibitor crizotinib, which down-regulated p-MET and CD73 levels in parallel (**Fig. 4E**). The impact of MET inhibition on CD73 suppression was also similar to that of MEK/ERK inhibition by trametinib, which has been linked to regulation of CD73 levels in melanoma (**Fig. 4E**; ref. 40). Finally, because H596 cells failed to express high levels of CD73 but lacked robust p-MET, we treated with HGF to stimulate MET activity, which resulted in an increase in CD73, whereas the simultaneous addition of crizotinib decreased the levels of p-MET and of CD73 (**Fig. 4F**). Taken together, these data provide strong additional evidence that oncogenic activation of *MET* and downstream signaling induces CD73 across lung cancer histologies.

CD73 is regulated by FRA1 in MET-amplified EGFR-TKI–resistant cells and generates adenosine

The top genetic dependencies in cancer cell lines with high levels of CD73, uncovered not only MET, but also the FOS like antigen 1 (*FOSL1*) as the top hit (**Fig. 3C**). We also noticed that *FOSL1* was among the top coexpressed genes with MET in our TCGA/CCLE expression analysis (**Fig. 3A**). FRA1 codes for a MEK-ERK–dependent

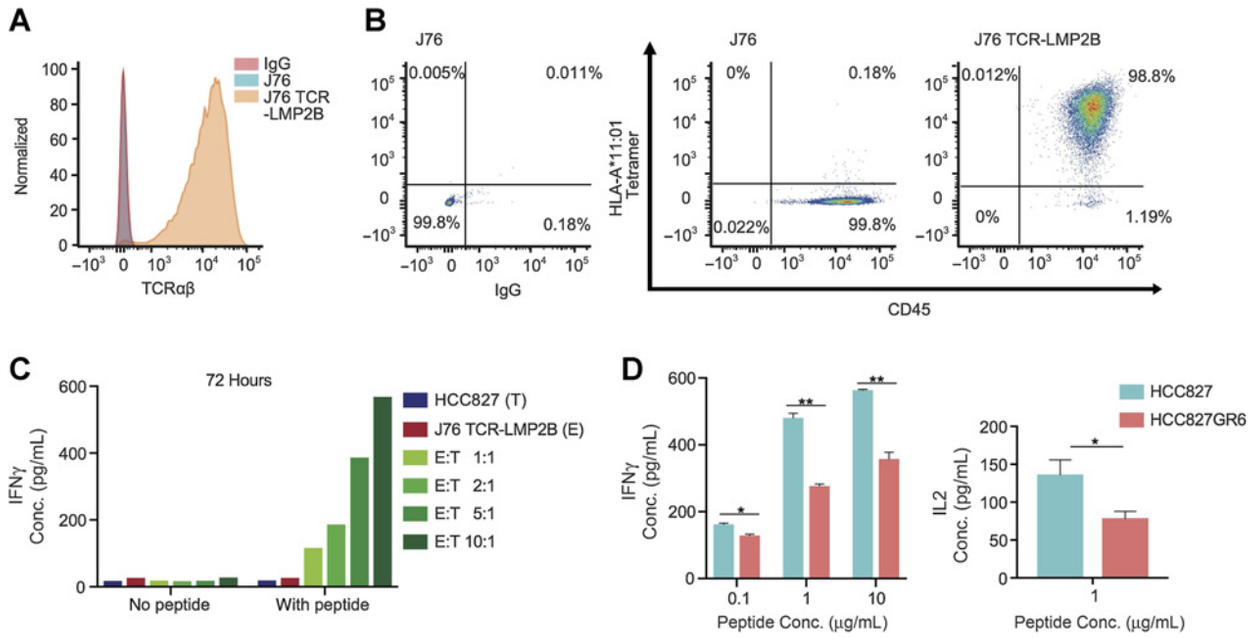


Figure 2.

Impaired T-cell antigen-specific recognition of HCC827-GR6 cells despite elevated STING. **A,** TCRαβ expression on J76 cells transduced with or without LMP2B TCR. **B,** HLA-A*11:01 Tetramer and CD45 expression on J76 cells transduced with or without LMP2B TCR. **C,** ELISA of human IFNγ levels in CM derived from HCC827 cells with J76 cells transduced with LMP2B TCR for 72 hours and pretreated ±10 μg/mL peptides (SSCSSCPLSK) for 3 hours. Each data represents one experiment. **D,** ELISA of human IFNγ levels in CM derived from HCC827 or HCC827GR6 cells with J76 cells transduced with LMP2B TCR for 72 hours and pretreated ± 0.1, 1, 10μg/mL peptides (SSCSSCPLSK) for 3 hours. Mean ± SEM of $n = 2$ biological replicates is shown (unpaired two-tailed Student *t* test; left). ELISA of human IL2 levels in CM derived from HCC827 or HCC827GR6 cells with J76 cells transduced with LMP2B TCR for 72 hours and pretreated ±1 μg/mL peptides (SSCSSCPLSK) for 3 hours. Mean ± SEM of $n = 3$ biological replicates shown (unpaired two-tailed Student *t* test; right). *, $P < 0.05$; **, $P < 0.005$.

transcription factor. Although it has not been previously implicated in regulation of CD73 in lung cancer, FRA1 is involved in MAPK signaling, suggesting its upregulation could provide a link between MET, MEK/ERK activation, and control of CD73 expression, which has been reported to drive CD73 expression in melanoma. Indeed, analysis of RPPA data in the CCLE revealed induction of FRA1 and MEK/ERK signaling as well as *MET* in *CD73* high lung cancer cell lines (Supplementary Fig. S3A). We also confirmed that levels of protein and RNA for FRA1 were both significantly upregulated in HCC827GR6 compared with HCC827 cells (**Fig. 5A**). In addition, MET deletion suppressed FRA1 expression in HCC827GR6 as well as DFCI-202 cells, supporting a direct functional relationship between MET activation and induction of FRA1 (**Fig. 5B**; Supplementary Fig. S3B).

To assess the direct relationship between FRA1 and regulation of CD73 expression in EGFR-mutated lung cancers, we next overexpressed FRA1 in HCC827 cells, which increased the expression of CD73 (**Fig. 5C**). We also deleted FRA1 using CRISPR/CAS9 in HCC827GR6 or DFCI-202 cells, which decreased CD73 levels (**Fig. 5D** and **E**; Supplementary Fig. S3C), confirming that FRA1 plays a key role to induce CD73 downstream of MET. In further consonance with this notion, whereas treatment of HCC827GR6 cells with savolitinib alone modestly diminished CD73 levels, trametinib treatment substantially downregulated CD73 expression (**Fig. 5F**), similar to what we observed in EBC1 cells (**Fig. 4E**). Thus, FRA1 and pERK coactivation downstream of MET induces CD73 in EGFR-TKI–resistant cells.

We next generated a protocol to evaluate the production of adenosine for a functional analysis of CD73 activity in this context (**Fig. 5G**). Adenosine levels in conditioned media were significantly reduced in HCC827GR6 cells following CD73 deletion, as compared with control cells (**Fig. 5H**). Conversely, production of adenosine was also significantly increased in HCC827 cells engineered to overexpress CD73 (**Fig. 5I**). Next, we compared levels of adenosine production between HCC827 and HCC827GR6 cells using this assay, and confirmed increased levels in HCC827GR6 cells, in consonance with their higher baseline levels of CD73 expression (**Fig. 5J**). Furthermore, adenosine production was decreased following MET deletion in HCC827GR6 cells with decreased CD73 (**Fig. 5K**). Taken together, these data confirm that MET signaling and FRA1 promote functional activation of CD73 in *EGFR*-mutated lung cancers.

Pemetrexed treatment co-opts cGAS-STING signaling in MET-amplified EGFR-TKI–resistant cells

We considered the possibility that co-opting this state by treating cells with existing lung cancer chemotherapeutic agents to drive STING pathway activation might overcome the immune suppressive consequences of CD73 activation. We first screened CXCL10 as an activation marker for STING, using multiple chemotherapeutic agents, following pulse treatment of HCC827GR6 cells (Supplementary Fig. S3D). Pemetrexed (PEM) and docetaxel (DOC) were the two most effective agents at increasing CXCL10 (**Fig. 6A**). We also confirmed that PEM treatment substantially activated TBK1-IRF3-STAT1 signaling along with CXCL10 in HCC827GR6 cells, but not in HCC827 and HCC827EPR cells, and that STING deletion suppressed the majority of this effect (**Fig. 6B** and **C**; Supplementary Fig. S3E and S3F). We also compared levels of downstream STAT1 activation following PEM versus DOC treatment across the different HCC827

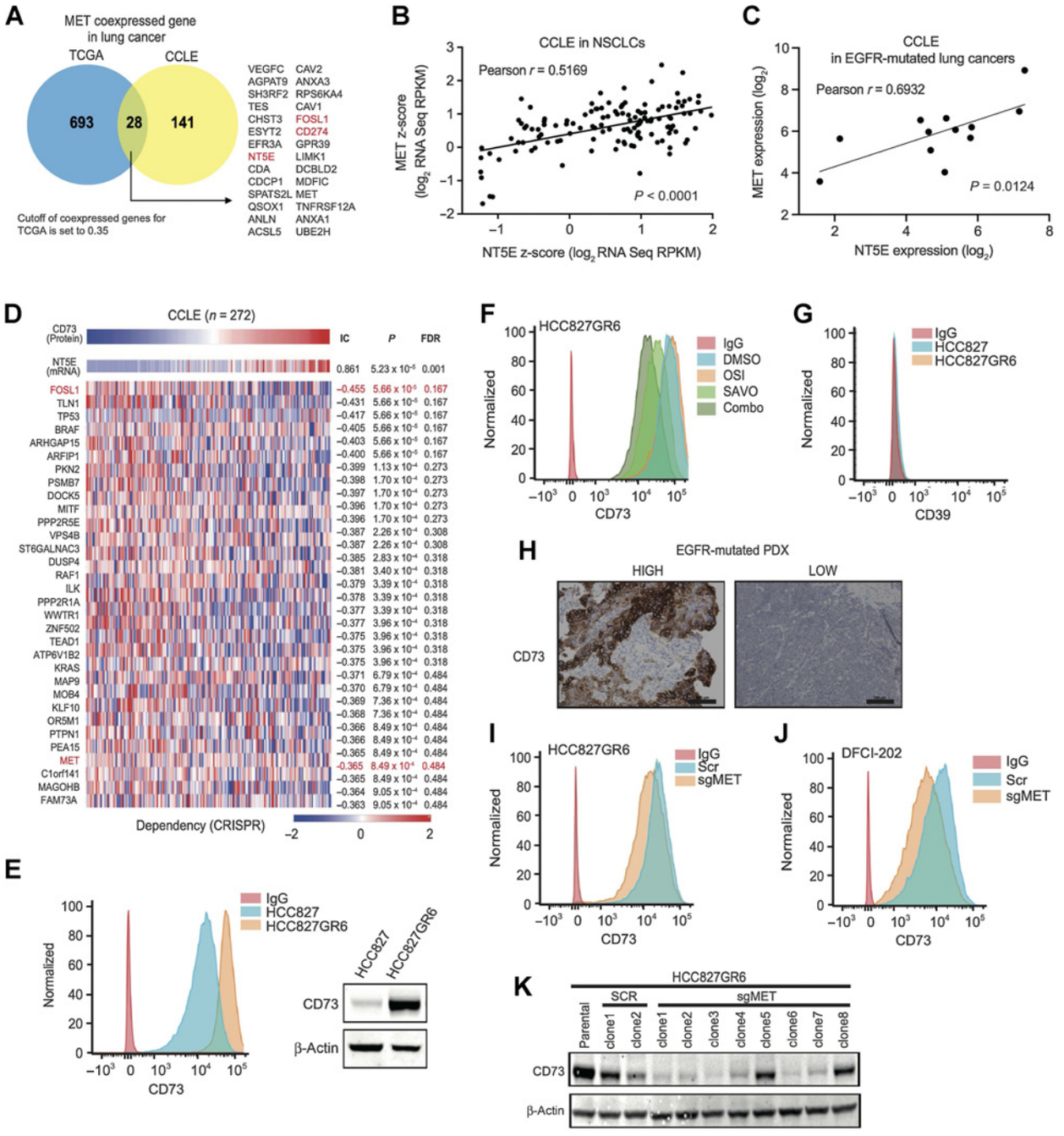


Figure 3.

CD73 coactivation with STING in MET-driven EGFR-TKI–resistant cells. **A,** Venn diagram showing the genes coexpressed with MET in NSCLCs in TCGA and CCLE. **B,** Correlation between *MET* and *NT5E* expression in lung cancer cell lines using CCLE. Pearson correlation and *P* value were calculated. **C,** Gene expression correlation between MET and NT5E in EGFR-mutated lung cancer cell lines using CCLE. *R* squared and *P* values were calculated using Pearson correlation. **D,** Top genetic dependencies identified on the basis of *NT5E* expression across CCLE and Depmap database ($n = 272$). The heatmap colors illustrate relative scores for mRNA and protein expression high (red) to low (blue), and the degree of dependency from high (blue) to low (red) as shown by the scale shown on the bottom right. **E,** CD73 expression on HCC827 and HCC827GR6 cells. **F,** CD73 expression on HCC827GR6 cells treated with 1 μmol/L osimertinib (OSI) and 1 μmol/L savolitinib (SAVO) for 48 hours. **G,** CD39 expression on HCC827 and HCC827GR6 cells. **H,** Representative CD73 IHC images of EGFR-mutated PDX samples. Representative image uses the same samples corresponding to the HIGH/LOW group of cMET in **Fig. 1E**. **I** and **J**, CD73 expression on HCC827GR6 or DFCI-202 cells with Scramble (Scr) or MET sgRNA. **K,** Correlation between MET and CD73 in HCC827GR6 signle-cell cloning with Scr or MET sgRNA.

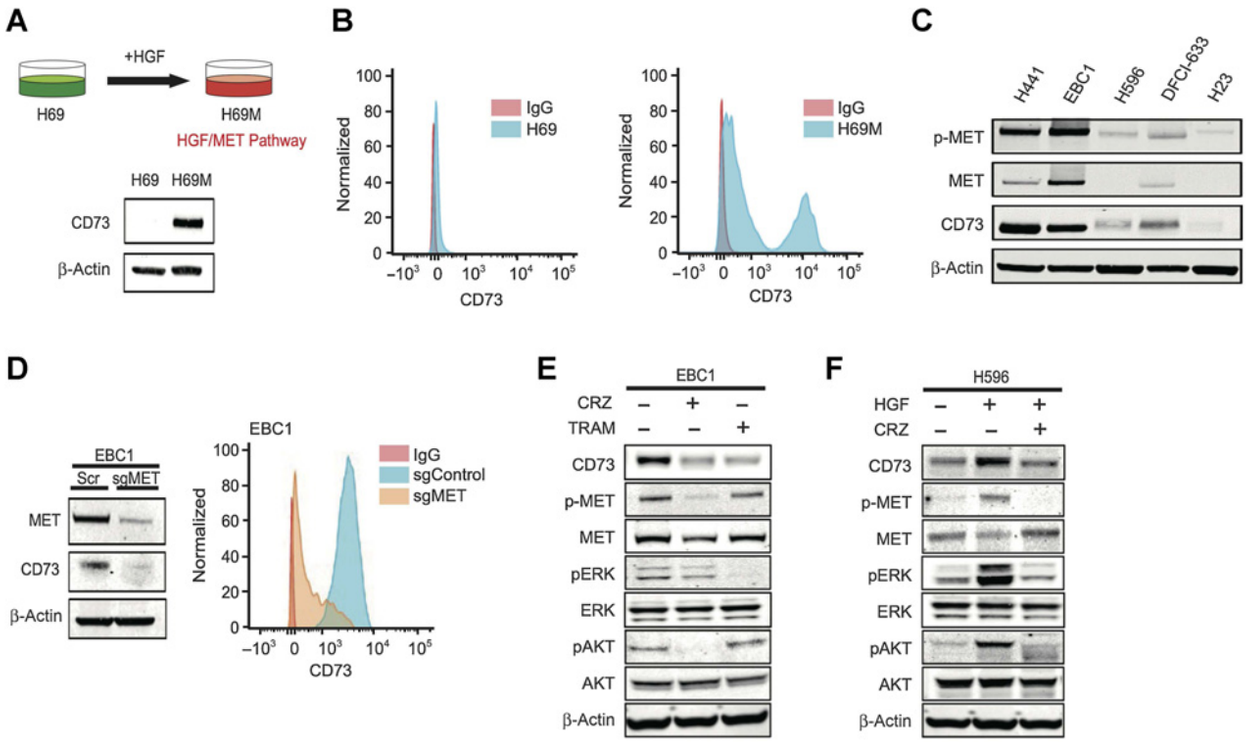


Figure 4.

CD73 is regulated by oncogenic *MET* in lung cancer cells. **A,** H69/H69M HGF-derived isogenic model and immunoblot of CD73 at basal expression. **B,** Flow cytometry of CD73 in H69M at basal condition. Red, isotype control. **C,** Immunoblot of different lung cancer cell lines used in the study. Protein basal status is indicated. **D,** Immunoblot showing the levels of MET and CD73 in the EBC1 cells after MET KO targeting MET (#1KO). Scr, scramble control. ACTIN, total protein-loading control (up). Flow cytometry assay of CD73 levels in the MET scr and KO EBC1 cells (down). **E,** Immunoblot of the indicated proteins in the cell lines after treatment with 0.1 μmol/L crizotinib (CRZ) and 30 nmol/L trametinib (TRAM). **F,** Immunoblot of the indicated proteins in the cell lines after treatment with 0.05 μmol/L HGF and 0.1 μmol/L crizotinib.

parental and EGFR-TKI–resistant models. Whereas DOC treatment induced STAT1 activation in all three cell lines, suggesting a nonspecific effect, PEM treatment induced STAT1 activation and CXCL10 secretion/mRNA expression only in HCC827GR6 cells (**Fig. 6D–F**). Furthermore, PEM treatment uniquely induced IFNB mRNA expression in HCC827GR6 cells (**Fig. 6G**). Finally, we observed that PEM was uniquely capable of inducing 2′-3′ cGAMP accumulation in HCC827GR6 cells (**Fig. 6H**), and that both PEM-induced STAT1 activation and 2′-3′ cGAMP accumulation was ablated by cGAS KO in HCC827GR6 cells (**Fig. 6I** and **J**). Taken together, PEM treatment co-opts endogenous cGAS-STING activation in MET-driven EGFR-TKI–resistant cell lines to promote TBK1-IRF3-STAT1 signaling, potentially enhancing the immunogenicity of this cell state.

PEM-induced immunogenicity is restrained by CD73

We therefore examined the implications of these findings on restoring immunogenicity using an allogeneic stem cell humanized xenograft mouse model as well as the TCR specific model we generated *in vitro*. Baseline HCC827GR6 CD73 KO xenograft tumor growth was significantly delayed in the humanized model as compared with control tumors ($P = 0.0276$; **Fig. 7A**). In contrast, we confirmed that CD73 deletion had no impact on cell proliferation rate *in vitro* (Supplementary Fig. S4A). We validated that circulating human $CD8^+$ T cells were established in the humanized model and found that the frequency of $CD8^+$ T cells infiltrating into tumors harvested at >1,000 mm^3 in size was significantly higher in CD73-deleted HCC827GR6 xenografts ($P = 0.0432$; **Fig. 7B**), and redistributed from the tumor periphery into the intratumoral compartment by CD8 IHC (**Fig. 7C** and **D**).

Next, we treated the remaining set of large tumors (>1,000 mm^3) with a single course of PEM, to assess the impact on tumor response in the setting of high tumor cell STING content (**Fig. 7E**). We observed dramatic reduction in tumor size in both CD73KO and control xenografted tumors (**Fig. 7F**). Because of complete tumor regressions, we were unable to quantify T-cell expansion within these tumors; however, reassessment from blood revealed a trend towards greater peripheral expansion of circulating CD8 T cells and dendritic cells in CD73 KO mice (Supplementary Fig. S4B–S4D). In consonance with this impressive impact of PEM treatment, we found that cell surface expression of HLA-A,B,C after treatment with PEM was specifically increased in HCC827GR6 cells, but not in HCC827 and HCC827EPR cells (**Fig. 7G** and **H**). We also confirmed that STING depletion specifically decreased baseline HLA-A,B,C expression in HCC827GR6 cells (Supplementary Fig. S5A) and that the increase in HLA-A,B,C associated with PEM treatment in HCC827GR6 cells was suppressed, albeit incompletely, by deleting STING (Supplementary Fig. S5B). These data confirm that PEM treatment is capable of restoring immunogenicity despite activation of CD73, though in the context of allogeneic T-cell response.

Finally, we examined the potential role of CD73 signaling in restraining this immunogenicity in the antigen specific model, given the artificial nature of humanized mouse models and the amplified

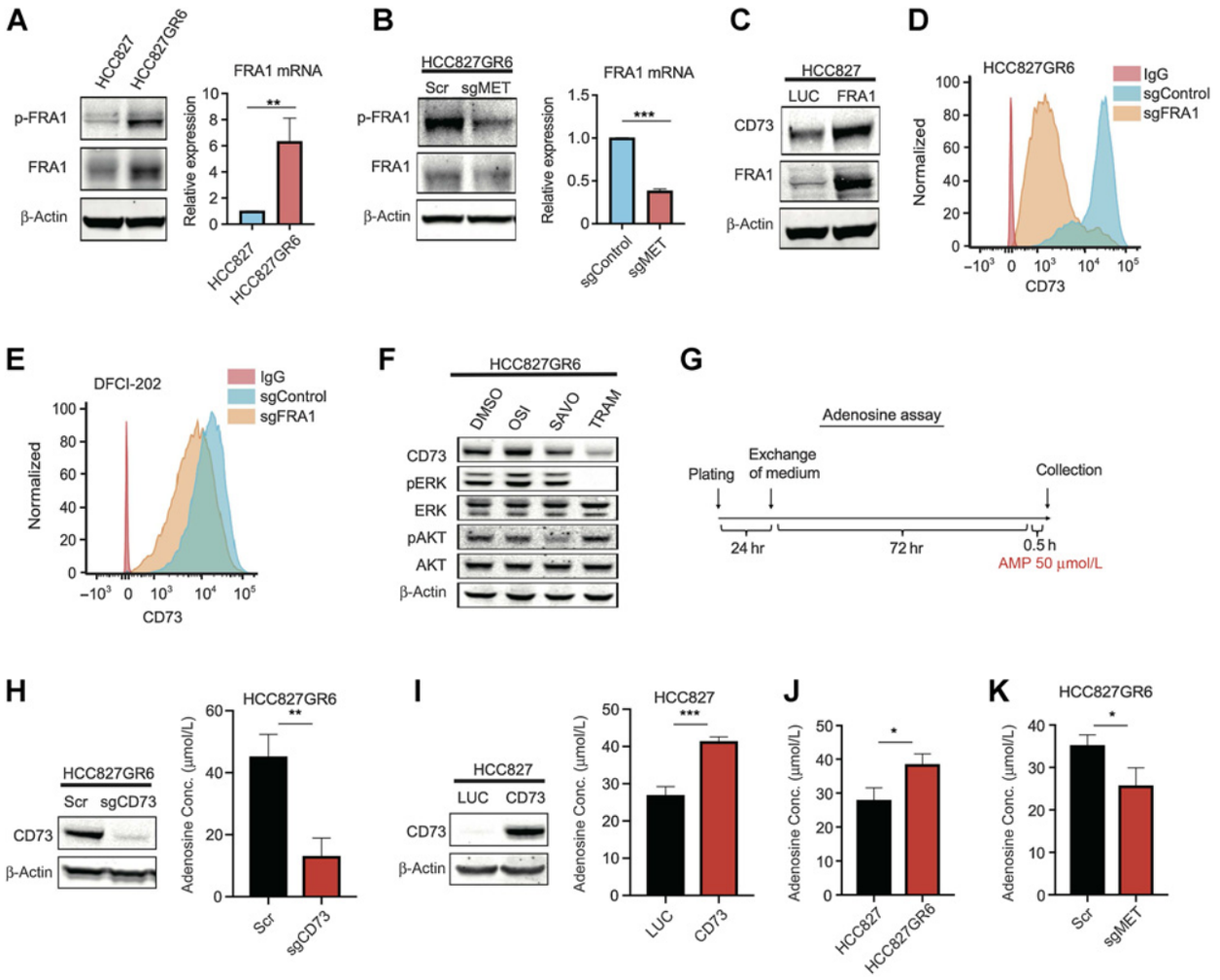


Figure 5.

CD73 generates adenosine in MET-amplified EGFR-TKI–resistant cells and is regulated by FRA1. **A** and **B,** Immunoblot of the phospho- and total FRA1 proteins or qRT-PCR of FRA1 in HCC827 and HCC827GR6 cell lines or HCC827GR6 cell lines with Scramble (Scr) or MET sgRNA. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **C,** Immunoblot of indicated proteins in HCC827 cells overexpressing LUC or FRA1. **D** and **E**, CD73 expression on HCC827GR6 or DFCI-202 cells with Scramble or FRA1 sgRNA. **F,** Immunoblot of indicated proteins HCC827GR6 cell lines treated with indicated compounds for 48 hours. OSI, osimertinib; SAVO, savolitinib; TRAM, trametinib. **G,** Adenosine assay protocol using Adenosine Assay Kit (Abcam, catalog no. ab211094). **H,** Immunoblot of indicated proteins in HCC827GR6 cells with Scramble or CD73 sgRNA. Adenosine levels in CM derived from HCC827GR6 cells with Scramble or CD73 sgRNA for 72 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **I,** Immunoblot of indicated proteins in HCC827 cells overexpressing LUC or CD73. Adenosine levels in CM derived from HCC827 cells overexpressing LUC or CD73 for 72 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **J,** Adenosine levels in CM derived from HCC827 or HCC827GR6 cells for 72 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **K,** Adenosine levels in CM derived from HCC827GR6 cells with Scramble or MET sgRNA for 72 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). *, $P < 0.05$; **, $P < 0.005$; ***, $P < 0.001$.

impact of allogenic reponse. Indeed, we found that PEM treatment further induced CD73 expression in HCC827-GR6 cells and DFCI-202 (**Fig 7I**; Supplementary Fig. S5C). Consistent with this observation, PEM enhanced basal production of adenosine, which was completely ablated by CD73 KO (**Fig. 7J**; Supplementary Fig. S5D and S5E). Using the antigen-specific J76 TCR-LMP2B coupled with specific recognition of the HLA A*11-associated SSCSSCPLSK peptide on HCC827GR6 cells, we confirmed that increasing the concentration of adenosine in this model decreased IFNγ production (Supplementary Fig. S5F). In this context, PEM treatment coupled with deletion of CD73 cooperated to promote TCR activation as measured by IFNγ and IL2 production (**Fig. 7K**). Furthermore, when AB 680 was used as a CD73 inhibitor instead of CRISPR CD73 KO in HCC827GR6 cells, the same combination effect with PEM was confirmed (Supplementary Fig. S5G). These data reveal that PEM treatment and CD73 inhibition together co-operate to enhance activation of specific T cells recognizing model antigens (**Fig. 7L**).

Discussion

Multiple strategies for targeting EGFR-TKI bypass pathways, including MET, have been examined, but to date limited durable activity has been observed in clinical practice (41–43). Here we report that *MET* amplification creates a broader cell state change in

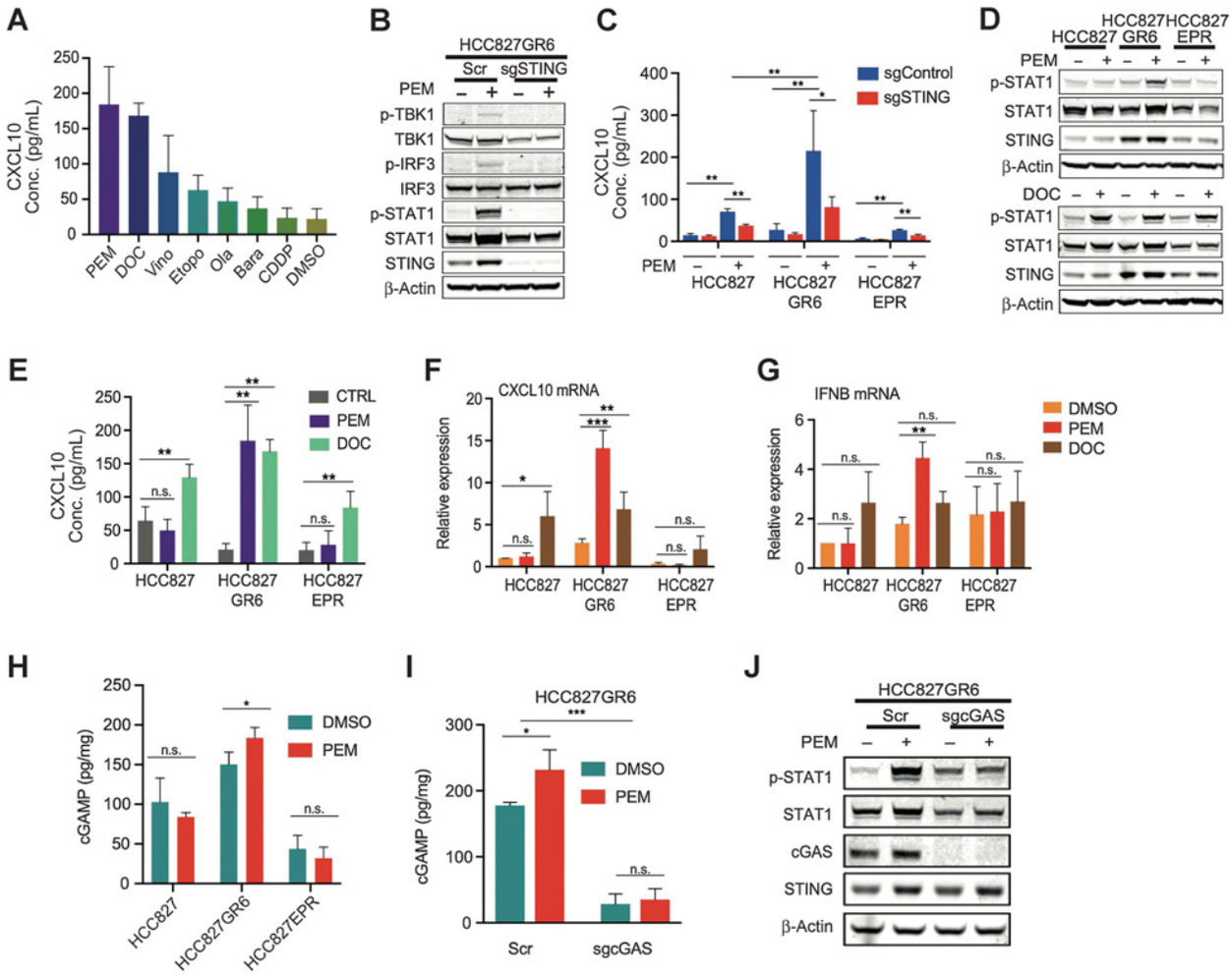


Figure 6.

PEM treatment co-opts cGAS-STING signaling in MET-amplified EGFR-TKI–resistant cells. **A,** ELISA of human CXCL10 levels in CM in HCC827GR6 cells treated with indicated compounds. **B,** Immunoblot of the indicated proteins in HCC827GR6 cells with Scramble (Scr) or STING sgRNA and treated ± PEM for 48 hours, followed by 24 hours of drug-free. **C,** ELISA of human CXCL10 levels in CM in HCC827, HCC827GR6, and HCC827EPR cell lines with Scramble or STING sgRNA and treated ± PEM for 48 hours, followed by 24 hours of drug-free. Mean ± SEM of $n = 4$ biological replicates is shown (unpaired two-tailed Student *t* test). **D** and **E,** Immunoblot of the indicated proteins or ELISA of human CXCL10 levels in CM in HCC827, HCC827GR6, and HCC827EPR cells treated ± PEM or DOC for 48 hours, followed by 24 hours of drug-free. **F** and **G,** qRT-PCR of human CXCL10 or human IFNB in HCC827, HCC827GR6, HCC827EPR cell lines treated ± PEM or DOC for 48 hours, followed by 24 hours of drug-free. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **H,** ELISA of human cGAMP levels in cell lysates derived from HCC827, HCC827GR6, HCC827EPR cells lines pretreated ± PEM for 48 hours, followed by 24 hours of drug-free. **I,** ELISA of human cGAMP levels in cell lysates derived from HCC827GR6 cells lines with Scramble or cGAS sgRNA pretreated ± PEM for 48 hours, followed by 24 hours of drug-free. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **J,** Immunoblot of indicated proteins in HCC827GR6 with Scramble or cGAS sgRNA pretreated ± PEM for 48 hours, followed by 24 hours of drug-free. *, $P < 0.05$; **, $P < 0.005$; ***, $P < 0.001$; n.s., nonsignificant.

EGFR-TKI–resistant NSCLC cells, which is associated with induction of STING and CD73, and could be leveraged for a novel therapeutic strategy. Although immune suppressive at baseline, co-opting STING activation via PEM treatment, while preventing generation of adenosine via CD73 inhibition, promotes an immunogenic cell state that facilitates antitumoral response. Indeed, PEM therapy in a humanized allogeneic mouse model results in robust tumor regressions, and when combined with CD73 KO significantly enhances TCR recognition of a model T-cell antigen.

Adenosine production downstream of the ectonucleotidases CD39 and CD73, creates an immune-suppressive tumor immune microenvironment and contributes to resistance to cancer immunotherapy (22, 27, 44). CD73 has been considered as an emerging immune checkpoint and a potential therapeutic target in the *EGFR*-mutant setting (23). Combining CD73 inhibition with other immunotherapies is currently under investigation in this context and in NSCLC more generally (NCT03822351, NCT03334617). Although we did not observe significant upregulation of PD-L1 in our models and *EGFR*-mutated lung cancer fails to respond to anti-PD(L)1 blockade (15, 16, 45), our results suggest that priming innate immune response in combination with CD73 inhibition might be more effective at triggering immunogenicity.

Our data also support the idea that cotargeting STING and CD73 might be more effective in *MET*-activated lung cancer cells. We previously linked epigenetic regulation of STING expression with HGF-MET activation and mesenchymal cell state (39), consistent

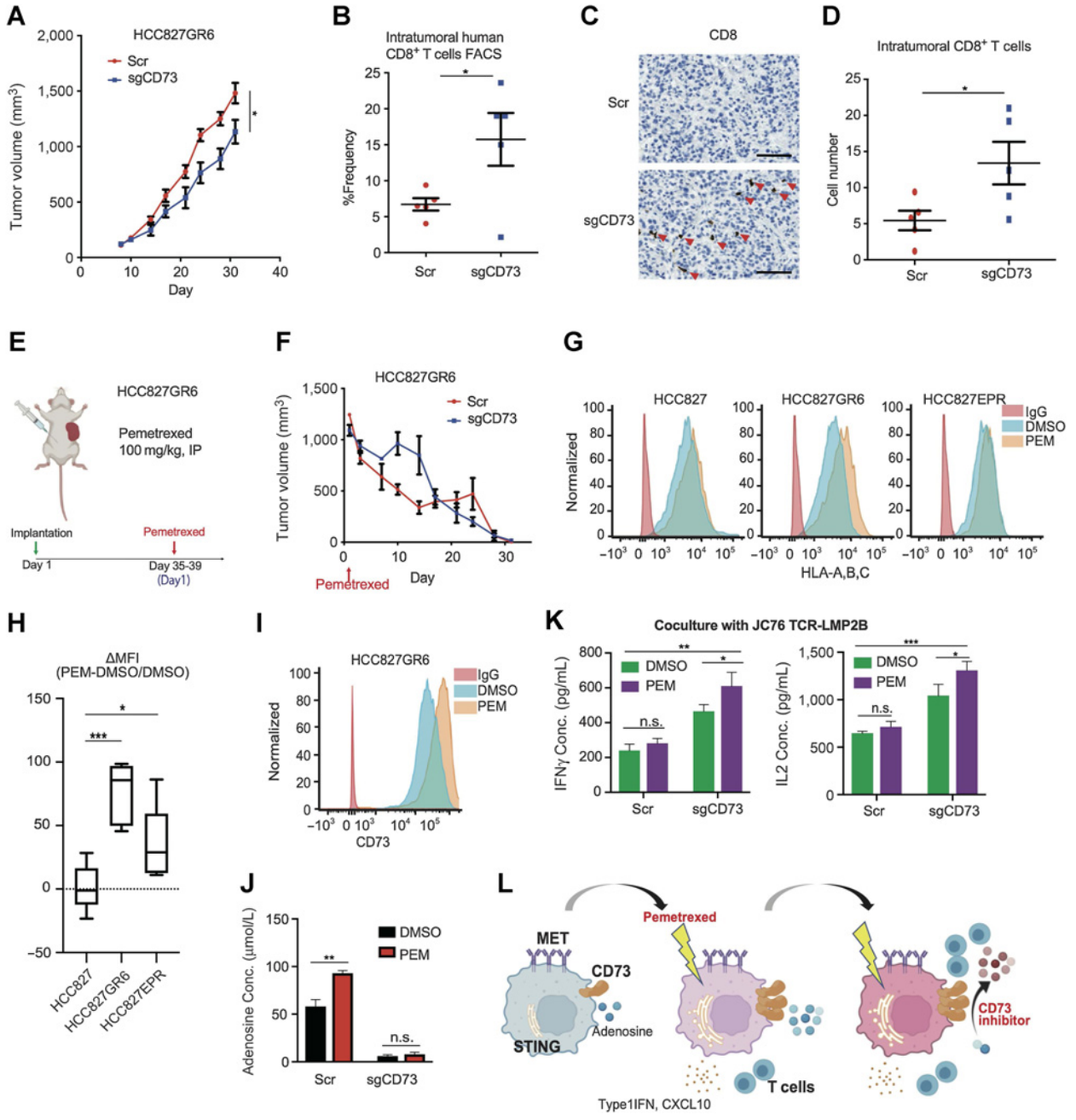


Figure 7.

PEM-induced immunogenicity is restrained by CD73. **A,** Tumor volume of HCC827GR6 cells with Scramble (Scr) or CD73sgRNA using humanized murine model ($n = 8$ in each group). **B,** Frequency of CD8$^+$ T cells infiltration among CD45$^+$ cells in mouse tumors ($n = 5$ in each group). **C** and **D**, Representative CD8 IHC images and analysis from Scramble or CD73sgRNA mouse tumors. Red arrows highlight intratumoral localization of CD8$^+$ T cells. Scale bar, 100 μmol/L. **E,** Schematic of study with PEM in humanized murine HCC827GR6 cells models. **F,** Tumor volume of HCC827GR6 cells with Scramble or CD73 sgRNA with PEM treatment in humanized murine model ($n = 3$ in each group). **G** and **H**, HLA-A,B,C expression on HCC827, HCC827GR6, and HCC827EPR cells treated ± PEM for 48 hours, followed by 24 hours of drug-free. MFI, mean fluorescence intensity. ΔMFI, (Pemetrexed-DMSO (dimethyl sulfoxide))/DMSO. Mean ± SEM of $n = 5$ biological replicates is shown (unpaired two-tailed Student *t* test). **I,** CD73 expression on HCC827GR6 cells treated ± PEM for 48 hours, followed by 24 hours of drug-free. **J,** Adenosine levels in CM derived from HCC827GR6 cell lines with Scramble or CD73 sgRNA treated ± PEM for 48 hours for 72 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **K,** ELISA of human IFNγ and IL2 levels in CM derived from HCC827GR6 cell lines with Scramble or CD73 sgRNA pretreated ± PEM for 48 hours with J76 cells transduced with LMP2B TCR for 72 hours and pretreated ±1 μg/mL peptides (SSCSSCPLSK) for 3 hours. Mean ± SEM of $n = 3$ biological replicates is shown (unpaired two-tailed Student *t* test). **L,** Schematic of treatment strategy on in MET-driven EGFR-TKI–resistant cells. *, $P < 0.05$; **, $P < 0.005$; ***, $P < 0.001$; n.s., nonsignificant.

with our finding that STING is uniquely upregulated in *MET*-amplified EGFR-TKI–resistant cells. Furthermore, we discover here that activation of *FOSL1* (FRA1) and MEK-ERK signaling in concert with MET simultaneously promotes CD73 expression. This finding is also reminiscent of similar work in melanoma that has implicated MAPK pathway activation and a mesenchymal cells state with CD73 upregulation in that context (40).

Importantly, we discover that PEM, the most frequently used cytotoxic drug in *EGFR*-mutated lung cancer (46) once they become resistant to targeted therapy, enhances immunogenicity of EGFR-mutated lung cancer by activating the STING pathway. This occurs most prominently in the context of *MET* amplification, increasing production of type I IFN, CXCL10, and HLA-A,B,C expression, but also generating adenosine. Recently, the phase II COAST clinical trial, evaluating the combination of oleclumab, a CD73 inhibitor, with durvalumab (anti-PDL1) has shown promising results after chemoradiation in locally advanced NSCLC (47). Because radiotherapy is another known activator of the cGAS-STING pathway (48), this early clinical observation may lend further credence to the idea that STING activation and CD73 inhibition is critical in promoting immunogenicity. It would be interesting to explore whether *MET* expression is linked to enhanced activity in this context. In addition, CD73 inhibition could potentially enhance immunogenicity of novel MET antibody drug conjugates, which have the potential to activate STING and are also being tested in EGFR-mutant *MET*-amplified NSCLC and in MET-activated NSCLC more broadly (5). Furthermore, MET amplification is also broadly found in cancers other than NSCLC, such as gastroesophageal carcinomas and the effect of combining CD73 inhibition with approaches that activate STING could be examined in these additional subgroups of carcinomas (49, 50).

Finally, multiple CD73 inhibitors in addition to oleclumab are in clinical trial development (e.g., NCT05431270, NCT04797468, NCT05143970, NCT04572152, and NCT03454451), and PEM is a standard chemotherapeutic option in lung adenocarcinoma. Clinical translation of this proof of principle therapeutic combination and determination of activity in *MET*-driven subsets of lung cancer should therefore be relatively straightforward.

Authors' Disclosures

S. Kitajima reports grants from Boehringer Ingelheim outside the submitted work. K. Haratani reports personal fees from AS ONE Corporation, Bristol-Myers Squibb Co. Ltd., Chugai Pharmaceutical Co. Ltd., Ono Pharmaceutical Co. Ltd.; grants and personal fees from AstraZeneca K.K. and MSD K.K. outside the submitted work. S.K. Sundararaman reports personal fees from LEK Consulting outside the submitted work. E.H. Knelson reports other support from Merck and grants from Takeda outside the submitted work. R. Uppaluri reports personal fees from Merck, Inc. outside the submitted work. P.C. Gokhale reports grants from Marengo Therapeutics, Epizyme Inc., Daiichi Sankyo, Foghorn Therapeutics, 28-7 Therapeutics, Moderna, and Kymera Therapeutics outside the submitted work. P.A. Jänne reports grants and personal fees from AstraZeneca during the conduct of the study; grants and personal fees from Boehringer Ingelheim, Eli Lilly, Daiichi Sankyo, Takeda Oncology; personal fees from Pfizer, Roche/Genentech, Chugai Pharmaceuticals, SFJ Pharmaceuticals, Voronoi, Biocartis, Novartis, Sanofi Oncology, Mirati Therapeutics, Transcenta, Silicon Therapeuticsons, Syndax, Nuvalent, Bayer, Esai, Allorion Therapeutics, Accutar Biootech, AbbVie; grants from Revolution Medicines, PUMA, and Astellas outside the submitted work; in addition, P.A. Jänne has a patent for EGFR mutations issued and licensed to LabCorp. D.A. Barbie reports grants from Loxo Oncology at Lilly during the conduct of the study; personal fees from Qiagen/N of One and Exo Therapeutics; grants from BMS, Gilead, Takeda, and Novartis; other support from Xsphera Biosciences outside the submitted work. No disclosures were reported by the other authors.

Authors' Contributions

R. Yoshida: Conceptualization, data curation, software, investigation, visualization, methodology, writing–original draft, writing–review and editing. **M. Saigi:** Conceptualization, data curation, supervision, funding acquisition, investigation, visualization, writing–original draft, project administration, writing–review and editing. **T. Tani:** Conceptualization, resources, data curation, formal analysis, supervision, investigation, visualization, methodology, writing–original draft, writing–review and editing. **B.F. Springer:** Data curation, investigation, visualization, writing–original draft, writing–review and editing. **H. Shibata:** Data curation, formal analysis, investigation, visualization, methodology. **S. Kitajima:** Data curation, investigation, methodology. **N.R. Mahadevan:** Formal analysis, investigation, methodology. **M. Campisi:** Investigation, methodology. **W. Kim:** Software, formal analysis, investigation, visualization, methodology. **Y. Kobayashi:** Investigation, methodology. **T.C. Thai:** Software, formal analysis, investigation, visualization, methodology. **K. Haratani:** Investigation, methodology. **Y. Yamamoto:** Investigation. **S.K. Sundararaman:** Investigation, methodology. **E.H. Knelson:** Resources, investigation. **A. Vajdi:** Formal analysis, investigation. **I. Canadas:** Resources, supervision, investigation. **R. Uppaluri:** Formal analysis, supervision, investigation. **C.P. Paweletz:** Supervision, investigation, methodology. **J.J. Miret:** Supervision, investigation. **P.H. Lizotte:** Investigation, methodology. **P.C. Gokhale:** Investigation, methodology. **P.A. Jänne:** Conceptualization, resources, supervision, investigation, methodology, writing–original draft, writing–review and editing. **D.A. Barbie:** Conceptualization, supervision, funding acquisition, investigation, visualization, writing–original draft, project administration, writing–review and editing.

Acknowledgments

This work was supported by NIH R01CA190294 (D.A. Barbie), a DFCI-Lilly Oncology Research Grant (D.A. Barbie), and the Candace Bagby fund for Lung Cancer Research (D.A. Barbie). M. Saigi received a SEOM grant and M. Saigi is currently supported by a Joan Rodes contract from the Instituto de Salud Carlos III (JR20/00015). The authors also thank Andrea Protti and Quang-De Nyugen for help with radiation of humanized mice.

The publication costs of this article were defrayed in part by the payment of publication fees. Therefore, and solely to indicate this fact, this article is hereby marked "advertisement" in accordance with 18 USC section 1734.

Note

Supplementary data for this article are available at Cancer Research Online (http://cancerres.aacrjournals.org/).

Received March 8, 2022; revised June 27, 2022; accepted August 29, 2022; published first September 6, 2022.

References

1. Kwon J, Bakhoum SF. The cytosolic DNA-sensing cGAS-STING pathway in cancer. Cancer Discov 2020;10:26–39.
2. Soria JC, Ohe Y, Vansteenkiste J, Reungwetwattana T, Chewaskulyong B, Lee KH, et al. Osimertinib in untreated EGFR-mutated advanced non-small-cell lung cancer. N Engl J Med 2018;378:113–25.
3. Le X, Puri S, Negrao MV, Nilsson MB, Robichaux J, Boyle T, et al. Landscape of EGFR-dependent and -independent resistance mechanisms to osimertinib and continuation therapy beyond progression in EGFR-mutant NSCLC. Clin Cancer Res 2018;24:6195–203.
4. Nagano T, Tachihara M, Nishimura Y. Mechanism of resistance to epidermal growth factor receptor-tyrosine kinase inhibitors and a potential treatment strategy. Cells 2018;7:212.
5. DaSilva JO, Yang K, Surriga O, Nittoli T, Kunz A, Franklin MC, et al. A biparatopic antibody-drug conjugate to treat MET-expressing cancers, including those that are unresponsive to MET pathway blockade. Mol Cancer Ther 2021;20:1966–76.
6. Recondo G, Che J, Jänne PA, Awad MM. Targeting MET dysregulation in cancer. Cancer Discov 2020;10:922–34.

7. Paik PK, Felip E, Veillon R, Sakai H, Cortot AB, Garassino MC, et al. Tepotinib in non-small-cell lung cancer with MET exon 14 skipping mutations. N Engl J Med 2020;383:931–43.
8. Wolf J, Seto T, Han JY, Reguart N, Garon EB, Groen HJM, et al. Capmatinib in MET exon 14-mutated or MET-amplified non-small-cell lung cancer. N Engl J Med 2020;383:944–57.
9. Sequist LV, Han JY, Ahn MJ, Cho BC, Yu H, Kim SW, et al. Osimertinib plus savolitinib in patients with EGFR mutation-positive, MET-amplified, non-small-cell lung cancer after progression on EGFR tyrosine kinase inhibitors: interim results from a multicentre, open-label, phase 1b study. Lancet Oncol 2020;21:373–86.
10. Oxnard GR, Yang JC, Yu H, Kim SW, Saka H, Horn L, et al. TATTON: a multi-arm, phase Ib trial of osimertinib combined with selumetinib, savolitinib, or durvalumab in EGFR-mutant lung cancer. Ann Oncol 2020;31:507–16.
11. Yang JJ, Fang J, Shu YQ, Chang JH, Chen GY, He JX, et al. A phase Ib study of the highly selective MET-TKI savolitinib plus gefitinib in patients with EGFR-mutated, MET-amplified advanced non-small-cell lung cancer. Invest New Drugs 2021;39:477–87.
12. Awad MM, Gadgeel SM, Borghaei H, Patnaik A, Yang JC, Powell SF, et al. Long-term overall survival from KEYNOTE-021 cohort G: pemetrexed and carboplatin with or without pembrolizumab as first-line therapy for advanced non-squamous NSCLC. J Thorac Oncol 2021;16:162–8.
13. Reck M, Rodríguez-Abreu D, Robinson AG, Hui R, Csőszi T, Fülöp A, et al. Pembrolizumab versus chemotherapy for PD-L1-positive non-small-cell lung cancer. N Engl J Med 2016;375:1823–33.
14. Lee CK, Man J, Lord S, Links M, Gebski V, Mok T, et al. Checkpoint inhibitors in metastatic EGFR-mutated non-small cell lung cancer-a meta-analysis. J Thorac Oncol 2017;12:403–7.
15. Gainor JF, Shaw AT, Sequist LV, Fu X, Azzoli CG, Piotrowska Z, et al. EGFR mutations and ALK rearrangements are associated with low response rates to PD-1 pathway blockade in non-small cell lung cancer: a retrospective analysis. Clin Cancer Res 2016;22:4585–93.
16. Offin M, Rizvi H, Tenet M, Ni A, Sanchez-Vega F, Li BT, et al. Tumor mutation burden and efficacy of EGFR-tyrosine kinase inhibitors in patients with EGFR-mutant lung cancers. Clin Cancer Res 2019;25:1063–9.
17. Saigi M, Alburquerque-Bejar JJ, Mc Leer-Florin A, Pereira C, Pros E, Romero OA, et al. MET-oncogenic and JAK2-inactivating alterations are independent factors that affect regulation of PD-L1 expression in lung cancer. Clin Cancer Res 2018;24:4579–87.
18. Lee HJ, Zhuang G, Cao Y, Du P, Kim HJ, Settleman J. Drug resistance via feedback activation of Stat3 in oncogene-addicted cancer cells. Cancer Cell 2014;26:207–21.
19. Blakely CM, Pazarentzos E, Olivas V, Asthana S, Yan JJ, Tan I, et al. NF-kappaB-activating complex engaged in response to EGFR oncogene inhibition drives tumor cell survival and residual disease in lung cancer. Cell Rep 2015;11:98–110.
20. Terai H, Kitajima S, Potter DS, Matsui Y, Quiceno LG, Chen T, et al. ER stress signaling promotes the survival of cancer "persister cells" tolerant to EGFR tyrosine kinase inhibitors. Cancer Res 2018;78:1044–57.
21. Gong K, Guo G, Panchani N, Bender ME, Gerber DE, Minna JD, et al. EGFR inhibition triggers an adaptive response by co-opting antiviral signaling pathways in lung cancer. Nature Cancer 2020;1:394–409.
22. Boison D, Yegutkin GG. Adenosine metabolism: emerging concepts for cancer therapy. Cancer Cell 2019;36:582–96.
23. Le X, Negrao MV, Reuben A, Federico L, Diao L, McGrail D, et al. Characterization of the immune landscape of EGFR-mutant NSCLC identifies CD73/adenosine pathway as a potential therapeutic target. J Thorac Oncol 2021;16:583–600.
24. Tu E, McGlinchey K, Wang J, Martin P, Ching SL, Floc'h N, et al. Anti-PD-L1 and anti-CD73 combination therapy promotes T cell response to EGFR-mutated NSCLC. JCI Insight 2022;7:e142843.
25. Allard B, Allard D, Buisseret L, Stagg J. The adenosine pathway in immuno-oncology. Nat Rev Clin Oncol 2020;17:611–29.
26. Deaglio S, Dwyer KM, Gao W, Friedman D, Usheva A, Erat A, et al. Adenosine generation catalyzed by CD39 and CD73 expressed on regulatory T cells mediates immune suppression. J Exp Med 2007;204:1257–65.
27. Vijayan D, Young A, Teng MWL, Smyth MJ. Targeting immunosuppressive adenosine in cancer. Nat Rev Cancer 2017;17:709–24.
28. Young A, Ngiow SF, Gao Y, Patch AM, Barkauskas DS, Messaoudene M, et al. A2AR adenosine signaling suppresses natural killer cell maturation in the tumor microenvironment. Cancer Res 2018;78:1003–16.
29. Inoue Y, Yoshimura K, Kurabe N, Kahyo T, Kawase A, Tanahashi M, et al. Prognostic impact of CD73 and A2A adenosine receptor expression in non-small-cell lung cancer. Oncotarget 2017;8:8738–51.
30. Turcotte M, Spring K, Pommey S, Chouinard G, Cousineau I, George J, et al. CD73 is associated with poor prognosis in high-grade serous ovarian cancer. Cancer Res 2015;75:4494–503.
31. Loi S, Pommey S, Haibe-Kains B, Beavis PA, Darcy PK, Smyth MJ, et al. CD73 promotes anthracycline resistance and poor prognosis in triple negative breast cancer. Proc Natl Acad Sci U S A 2013;110:11091–6.
32. Rocha P, Salazar R, Zhang J, Ledesma D, Solorzano JL, Mino B, et al. CD73 expression defines immune, molecular, and clinicopathological subgroups of lung adenocarcinoma. Cancer Immunol Immunother 2021;70:1965–76.
33. Kitajima S, Asahina H, Chen T, Guo S, Quiceno LG, Cavanaugh JD, et al. Overcoming resistance to dual innate immune and MEK inhibition downstream of KRAS. Cancer Cell 2018;34:439–52.e6.
34. Pacini C, Dempster JM, Boyle I, Gonçalves E, Najgebauer H, Karakoc E, et al. Integrated cross-study datasets of genetic dependencies in cancer. Nat Commun 2021;12:1661.
35. Falahat R, Berglund A, Putney RM, Perez-Villarroel P, Aoyama S, Pilon-Thomas S, et al. Epigenetic reprogramming of tumor cell-intrinsic STING function sculpts antigenicity and T cell recognition of melanoma. Proc Natl Acad Sci U S A 2021;118:e2013598118.
36. Zheng Y, Parsonage G, Zhuang X, Machado LR, James CH, Salman A, et al. Human leukocyte antigen (HLA) A*1101-restricted epstein-barr virus-specific T-cell receptor gene transfer to target nasopharyngeal carcinoma. Cancer Immunol Res 2015;3:1138–47.
37. Rosskopf S, Leitner J, Paster W, Morton LT, Hagedoorn RS, Steinberger P, et al. A Jurkat 76 based triple parameter reporter system to evaluate TCR functions and adoptive T cell strategies. Oncotarget 2018;9:17608–19.
38. Cañadas I, Rojo F, Taus Á, Arpí O, Arumí-Uría M, Pijuan L, et al. Targeting epithelial-to-mesenchymal transition with Met inhibitors reverts chemoresistance in small cell lung cancer. Clin Cancer Res 2014;20:938–50.
39. Cañadas I, Thummalapalli R, Kim JW, Kitajima S, Jenkins RW, Christensen CL, et al. Tumor innate immunity primed by specific interferon-stimulated endogenous retroviruses. Nat Med 2018;24:1143–50.
40. Reinhardt J, Landsberg J, Schmid-Burgk JL, Ramis BB, Bald T, Glodde N, et al. MAPK signaling and inflammation link melanoma phenotype switching to induction of CD73 during immunotherapy. Cancer Res 2017;77:4697–709.
41. Piotrowska Z, Isozaki H, Lennerz JK, Gainor JF, Lennes IT, Zhu VW, et al. Landscape of acquired resistance to osimertinib in EGFR-mutant NSCLC and clinical validation of combined EGFR and RET inhibition with osimertinib and BLU-667 for acquired RET fusion. Cancer Discov 2018;8:1529–39.
42. Bahcall M, Sim T, Paweletz CP, Patel JD, Alden RS, Kuang Y, et al. Acquired METD1228V mutation and resistance to MET inhibition in lung cancer. Cancer Discov 2016;6:1334–41.
43. Tricker EM, Xu C, Uddin S, Capelletti M, Ercan D, Ogino A, et al. Combined EGFR/MEK inhibition prevents the emergence of resistance in EGFR-mutant lung cancer. Cancer Discov 2015;5:960–71.
44. Leone RD, Emens LA. Targeting adenosine for cancer immunotherapy. J Immunother Cancer 2018;6:57.
45. Dong ZY, Zhang JT, Liu SY, Su J, Zhang C, Xie Z, et al. EGFR mutation correlates with uninflamed phenotype and weak immunogenicity, causing impaired response to PD-1 blockade in non-small cell lung cancer. Oncoimmunology 2017;6:e1356145.
46. Papadimitrakopoulou VA, Mok TS, Han JY, Ahn MJ, Delmonte A, Ramalingam SS, et al. Osimertinib versus platinum-pemetrexed for patients with EGFR T790M advanced NSCLC and progression on a prior EGFR-tyrosine kinase inhibitor: AURA3 overall survival analysis. Ann Oncol 2020;31:1536–44.
47. Herbst RS, Majem M, Barlesi F, Carcereny E, Chu Q, Monnet I, et al. COAST: an open-label, phase II, multidrug platform study of durvalumab alone or in combination with oleclumab or monalizumab in patients with unresectable, stage III non-small-cell lung cancer. J Clin Oncol 2022:Jco2200227.
48. Deng L, Liang H, Xu M, Yang X, Burnette B, Arina A, et al. STING-dependent cytosolic DNA sensing promotes radiation-induced type I interferon-dependent antitumor immunity in immunogenic tumors. Immunity 2014;41:843–52.
49. Jardim DL, Tang C, Gagliato Dde M, Falchook GS, Hess K, Janku F, et al. Analysis of 1,115 patients tested for MET amplification and therapy response in the MD anderson phase I clinic. Clin Cancer Res 2014;20:6336–45.
50. Smolen GA, Sordella R, Muir B, Mohapatra G, Barmettler A, Archibald H, et al. Amplification of MET may identify a subset of cancers with extreme sensitivity to the selective tyrosine kinase inhibitor PHA-665752. Proc Natl Acad Sci U S A 2006;103:2316–21.